

Session 1

Multi-Source Data Fusion

THE 92% SYSTEM IS INVALID

SESSION OBJECTIVE

Five defective data sources will be converted into one audit-ready AI modeling table.

840 unique records

0 duplicate keys

0 missing model cells

**2 future fields
quarantined**

PRIMARY ARTIFACT

AI Data Fusion Inspector

OPENING CHALLENGE

SYSTEM A

- Accuracy: 80%
- Future-only data: NO
- Duplicate audit: PASS
- Safe join contract: PASS

SYSTEM B

- Accuracy: 92%
- Future-only data: YES
- Duplicate audit: UNKNOWN
- Safe join contract: UNKNOWN

SYSTEM A is deployable. SYSTEM B is invalid despite the stronger score.

THE FIRST PRINCIPLE

The model is not the AI system.

The complete decision chain starts with what one row means and what information is available at prediction time.

SYSTEM MAP



Session 1 stops before model training. The artifact is a trusted modeling table.

FIVE DISCONNECTED SOURCES

WEATHER

temperature_c, rainfall_mm, humidity_pct

SOIL

soil_moisture, soil_ph, vegetation_index

SURVEILLANCE

disease_cases, pest_reports

REGION PROFILE

irrigation_capacity, vulnerability_index

OUTCOMES

risk_label plus controlled leakage fields

DEFINE THE OBSERVATION

ONE REGION x ONE MONTH = ONE ROW

The composite key is region_id + year + month.

EXPECTED GRAIN

10

regions

R01 to R10

7

years

2019 to 2025

12

months

calendar cycle

840

observations

10 x 7 x 12

Any operational source with 841 region-month rows deserves investigation before a merge.

SOURCE AUDIT - BEFORE CLEANING

Source	Rows	Missing cells	Duplicate keys
weather	841	1	1
soil	840	1	0
disease	841	1	1
regions	10	0	0
outcomes	840	0	0

The audit makes the 841-row anomalies visible before any model training.

INVESTIGATION MOMENT

Expected: 840. Weather: 841. Disease: 841.

The first hypothesis is a duplicate composite key - not 'more data'.

COUNT DUPLICATE COMPOSITE KEYS

```
KEYS = ["region_id", "year", "month"]  
  
duplicate_count = weather.duplicated(KEYS).sum()  
print("Duplicate weather keys:", duplicate_count)  
  
assert duplicate_count == 1
```

The checkpoint fails immediately if the controlled defect is not found.

BREAK IT

Run a strict one-to-one merge before cleaning.

The failure is expected. The merge contract should block silent row multiplication.

EXECUTABLE MERGE CONTRACT

```
weather.merge(  
    soil,  
    on=KEYS,  
    validate="one_to_one",  
)
```

The merge raises because weather still contains a duplicated region-month key.

INVALID IS NOT LOW

-18 mm rainfall is impossible, not an extreme low value.

Range violations are converted to missing values before interpolation.

CLEANING ORDER



The order is explicit. An impossible value must not participate in interpolation.

RANGE INVALIDATION

```
for column, (low, high) in VALID_RANGES.items():  
    if column in cleaned:  
        cleaned.loc[  
            ~cleaned[column].between(low, high),  
            column,  
        ] = np.nan
```

The invalid measurement becomes missing evidence before it is repaired.

REGION-AWARE INTERPOLATION

```
frame[column] = frame.groupby("region_id")[column].transform(  
    lambda series: series.interpolate(limit_direction="both")  
)  
  
frame[column] = frame[column].fillna(frame[column].median())
```

Interpolation stays inside the same region. Global median is only a final fallback.

MERGE CARDINALITY

one_to_one

weather + soil + surveillance + outcomes

many_to_one

monthly observations -> one region profile

never silent

cardinality is declared in every merge

SAFE FUSION

```
fused = weather_clean.merge(soil_clean, on=KEYS, validate="one_to_one")
fused = fused.merge(disease_clean, on=KEYS, validate="one_to_one")
fused = fused.merge(regions, on="region_id", validate="many_to_one")
fused = fused.merge(outcomes, on=KEYS, validate="one_to_one")
```

The row grain is protected at each boundary.

SIX TRANSPARENT FEATURES

drought_index

1 - rainfall / 115, clipped to [0,1]

heat_stress

scaled temperature above 27 C

disease_pressure

scaled disease cases

water_balance

soil + irrigation + drought composite

season_sin

cyclic month component

season_cos

cyclic month component

WHY CYCLIC MONTH?

December and January are neighbors.

Raw month numbers place them 11 units apart. Sine/cosine coordinates preserve the circular calendar.

CYCLIC MONTH FEATURES

```
angle = 2.0 * np.pi * (month - 1) / 12.0  
season_sin = np.sin(angle)  
season_cos = np.cos(angle)
```

Month becomes a point on a circle instead of a linear integer.

FEATURE REGISTRY ROLES

APPROVED

available at prediction time and accepted for modeling

PROHIBITED

post-outcome fields retained only for the leakage experiment

TARGET

risk_label

CONTEXT

identifiers and descriptive fields

LEAKAGE QUARANTINE

**future_loss_index and response_action_code
remain in the table - but never enter
MODEL_FEATURES.**

Retaining them makes the Session 2 failure reproducible while the registry makes the prohibition explicit.

AI DATA FUSION INSPECTOR - EXPECTED OUTPUT

```
Expected observations      : 840
Final observations        : 840
Duplicate keys            : 0
Missing model cells      : 0
Approved model features   : 16
Prohibited future fields  : 2
Fusion status             : PASSED
```

The notebook exports modeling_table.csv, source_audit.csv, and feature_registry.csv.

LIVE WORKSPACE CHECKPOINTS

1

Predict the reason for 841 rows before counting duplicate keys.

2

Trigger the strict merge failure before applying cleaning.

3

Detect the impossible rainfall value with a Boolean range mask.

4

Repeat the strict merge after cleaning and verify exactly 840 rows.

Build the six engineered features directly in the notebook.

Every checkpoint has a matching assertion in the complete reference notebook.

6

Create the feature registry and assert that prohibited fields are excluded from `MODEL_FEATURES`.

MINI-PROJECT

INPUT

five disconnected CSV sources

PROCESS

audit -> clean -> validate joins -> engineer
-> register

OUTPUT

trusted table + audit + feature registry

EVIDENCE TO RECORD

1

Which defects were found in each source?

2

Why did the strict merge fail before cleaning?

3

Why is invalidation performed before interpolation?

4

Why do December and January need cyclic encoding?

Why are post outcome fields retained but quarantined?

The evidence record becomes the opening context for Session 2.

SESSION RESULT

A trusted 840-row modeling table now exists.

The next question is whether a model can generalize to future data without cheating.